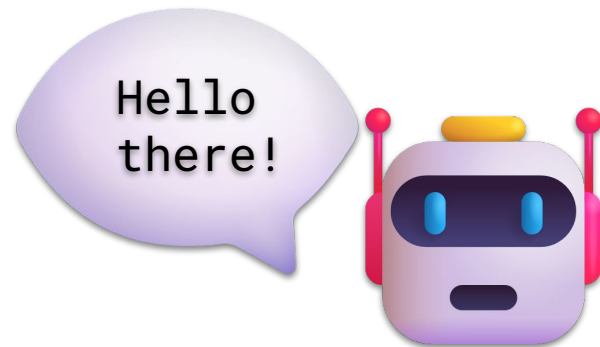




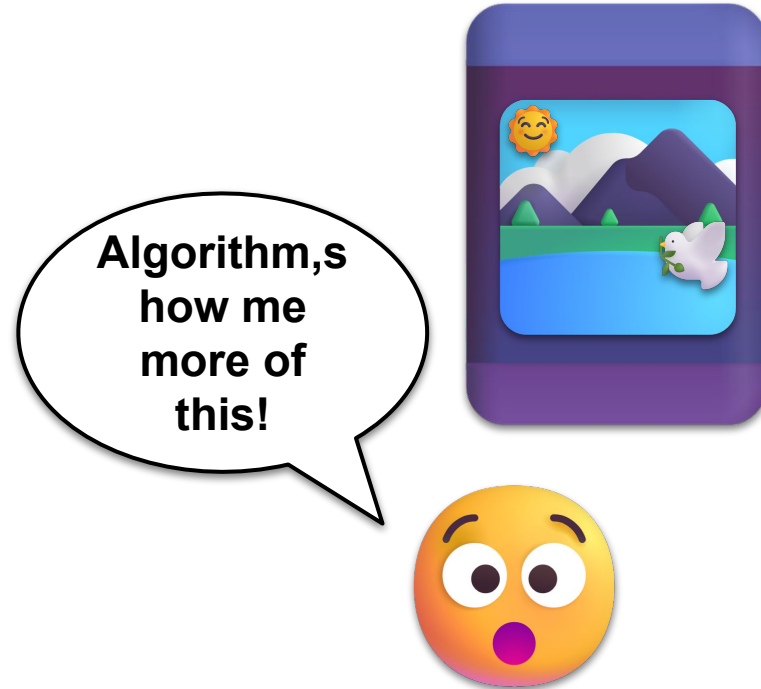
Algorithmic imaginaries

definition and future directions

Dr Ellenrose Firth
Dr Andreas Schellewald

Algorithms and media audiences

- **Algorithmic systems** play an increasingly central role in how audience interact with content and one another online (e.g., Das an Ytre-Arne 2018, Livingstone 2019, Kant 2020, Webster 2014).
- Users perceive “**the algorithm**” as a singular entity that they interact with, gets their feelings, they (attempt to) communicate with, and receive at times messages from (e.g., Bucher 2017, Kanthawala 2025, Siles 2023).



(user scenario and comment from Instagram Reels platform)

Many concepts and theories



Mental models of algorithms...

“subjective knowledge representations of technological systems ... making predictions about the outcome of the system ... typically incomplete, inaccurate, and may contain areas of uncertainty.” (Ngo et al. 2020)



Folk theories of algorithms...

“non-authoritative conceptions of the world that develop among non-professionals and circulate informally.” (Eslami et al. 2017)

“intuitive, informal theories ... to explain the outcomes, effects, or consequences of technological systems, which guide .. behavior towards said systems ... and include opinions and attitudes about possible consequences.” (DeVito et al. 2017)

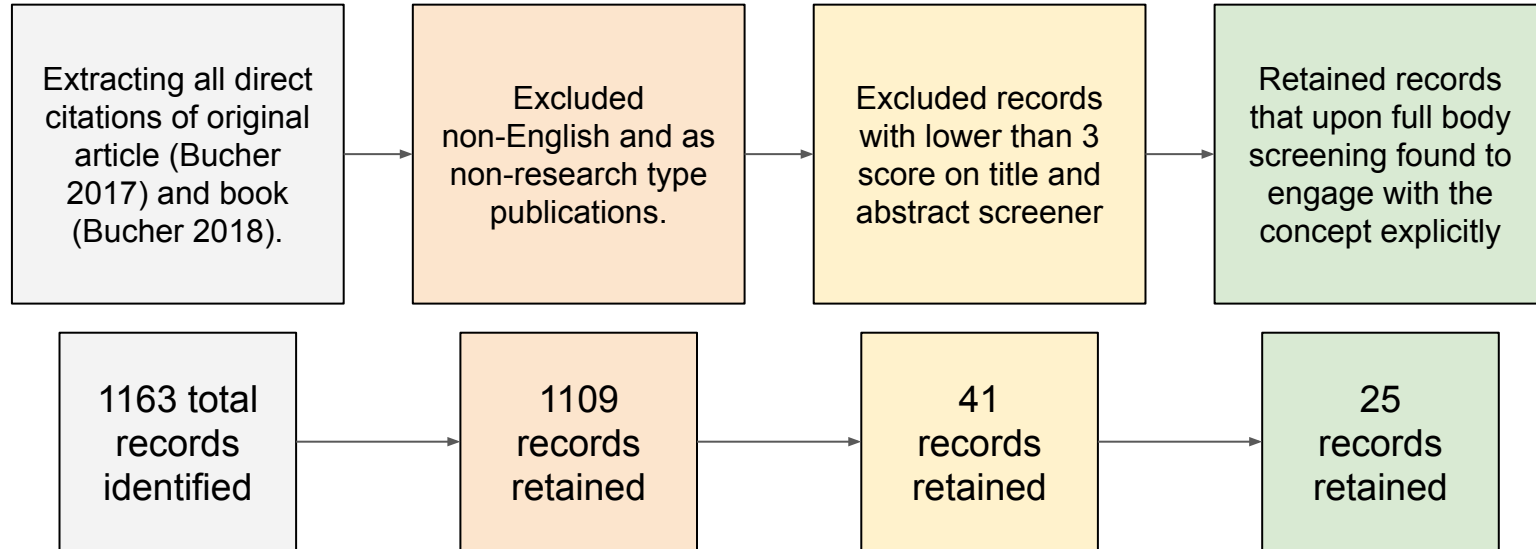


Imaginaries of algorithms...

“the space where algorithms and users meet ... the way in which people imagine, perceive and experience algorithms and what these imaginations make possible ... understood as mood and intensity corresponding to ‘forces of encounter’” (Bucher 2017)

Literature review

RQ: How has the algorithmic imaginaries concept been explicitly applied and developed since it was initially introduced?



Overview of findings

- The initial publications have been widely cited - and the work had a strong **sensitizing nature**.
- Even when the algorithmic imaginaries concept was not explicitly mentioned, the publications were referenced to underline **how people think and feel about algorithms**.
- Our analysis of the final body of records revealed **8 themes in which the concept has been discussed** and further developed.

- ❑ Theoretical (11): advance or rethink the concept of algorithmic imaginaries.
- ❑ **Governmental (8): Discuss governance and power in relation to algorithmic imaginaries.**
- ❑ Contextual (8): Explore algorithmic imaginaries in different contexts.
- ❑ **Collective (6): Focus on imaginaries formed within groups or communities.**
- ❑ Affective (6): emotional aspects shaping perceptions of algorithms.
- ❑ **Cultural (4): Examine cultural values and beliefs in algorithmic imaginaries.**
- ❑ Agentic (3): human agency in interacting with algorithms.
- ❑ Technical (2): technical contributions / design applications of the concept.

Governmental

- Building on Bucher's original work, a dominant theme has been the exploration of questions of power in algorithmic imaginaries.
- Imaginaries are positioned as contested forces - shaped by design, branding, advertising, as well as actively negotiated as shared affect and struggle in relation to powerful algorithmic orderings of relations.
- **!?** **Critical question:** what are the modes of power at play in how algorithms are imagined?



Creator-audience interactions that are mobilised by shared sentiments of being wrongly positioned by the algorithm (Jones 2023).

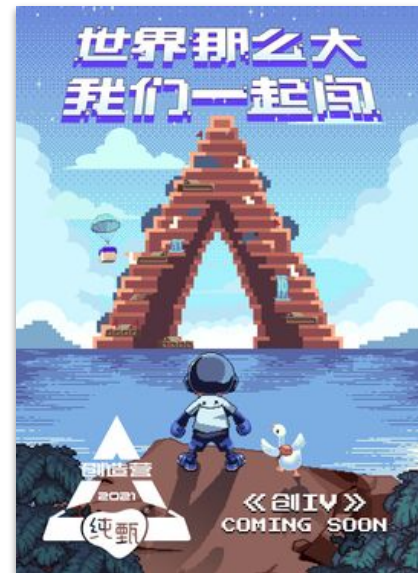


Amazon Alexa promoted with a sense of ordinariness, positive feelings, and gendered notions of feminized care (Kopitz 2021).

Collective

- Imaginaries are shaped not just by personal but also shared encounters with and sentiments of algorithmic systems and their actions.
- Imaginaries constitute a figurative space where users feel they are part of a larger group of people affected in the same way having the same experience.
- **!?** **Critical question:** how are imaginaries of algorithms formed through collectively experienced emotions in algorithmic spaces rather than individual affective force encounters with systems?

“He represents us ... every ‘no-background’ underdog who has a dream but never got a chance to achieve it. Now we can achieve it, collectively, through data.”
(Participant quoted in Zhang et al. 2023: 327)



Case of the "spiritual algorithmic imaginary", shaping not just how people make sense of recommended content, but also how they connect with likeminded audience members on the platform (Reinis and Laughlin 2025).

Cultural

- A number of studies show how audiences, but also other types of users, express and articulate their values when they imagine the algorithm.
- Building on the affective encounter, ways of imagining and taking a stand in the encounter are driven by values, norms, and beliefs.
- **!?** **Critical question:** what role do cultural norms, values, and beliefs play in the imaginations of algorithms? And how do algorithms, through the imaginary, potentially shape values and beliefs people hold?

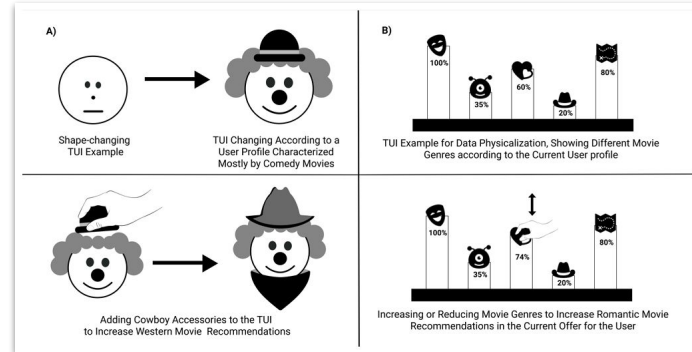


Case of BookTok and how the the ways in which users imagine the TikTok algorithm surfaces values and beliefs about literacy (Low et al. 2023)

Case of “Mutual aid” content on TiKTok and how idea about fairness, community, and solidarity are reflected in how algorithms are imagined and encountered (Maris et al. 2024)

Technical

- Research from the Human-Computer Interaction field showcasing the application of the concept in a technical context system design context.
- Building on Bucher's argument that users actively experiment with algorithms, the design visions showcase tangible and platform interfaces that add layers to how users can sense and control their algorithmic experiences.
- !?** **Critical questions:** how can design approaches be used to experimentally study the dynamics of imagining and experiencing algorithms - to envision better and fairer algorithmic futures?



Alvarado et al. 2021

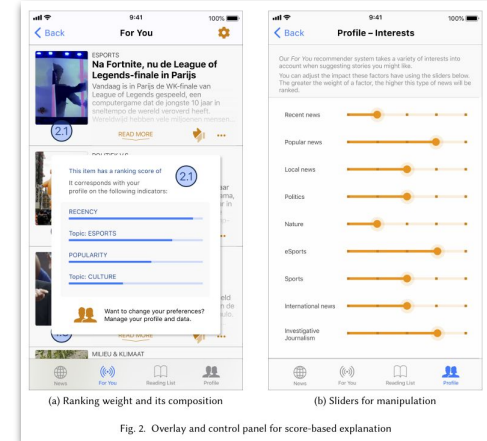


Fig. 2. Overlay and control panel for score-based explanation

Storms et al. 2022

Imaginaries and audiences

- **Expanded definition** foregrounding how imaginaries constitute playing fields where audiences and algorithms are socio-culturally configured as interaction partners.
- Parallel to the idea of **audiencing** (Fiske 1992) - focussing on the process of collectively imagining algorithms for audiences to produce “their own sense of their social identities and social relations” (Fiske 1992: 354).
- Imagining as a form of **significant activity** (Silverstone 1994). Recognizing that audiences are always in some way interactive with algorithms, imagination stands out as significant - allowing to develop “a theoretically motivated account of the dynamics of ... [algorithmic media] in everyday life” (Silverstone 1994: 158).



Algorithmic imaginaries as...

the space where algorithm and audiences are culturally situated in an affectively charged encounter, shaped by the socio-historically specific ways of designing, thinking about, and emotionally relating to data-driven and algorithmic systems.

Future directions

- The concept itself occupies a unique position in the literature due its sensitizing rather than definitive nature (Blumer 1954).
- Critical questions for future empirical understanding of algorithmic imaginaries
 - How do platforms and media producers seek to influence how audiences feel about and imagine their algorithms - in design, branding, and advertising?
 - What are the social and collective dynamics of how algorithms are imagined socially rather than individually?
 - What role do cultural norms, values, and beliefs play in positioning audiences and algorithms in the space of the imaginary?
- Exploring the role of imaginaries in the ever more personal and direct interactions with algorithms and chatbots powered by artificial intelligence technologies.

Thanks!

Ellenrose Firth
ellenrose.firth@uniroma1.it

Andreas Schellewald
aschellewald@acm.org